

Formal Verification meets AI

Using Lean as a Practical Bridge

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Core idea

Formal methods from this course apply directly to **verifying AI systems**. Lean 4 provides a practical bridge between theory and practice.

Roadmap:

- An AI failure that testing could not prevent
- Testing vs. formal proof — the fundamental gap
- Three verified Lean 4 properties of neural network components
- How each example connects to course concepts
- Where the field is heading: LeanDojo, AlphaProof, DeepSeek-Prover-V2

A Note on Tools

Retrieval (Claude + RAG)

Lecture PDFs 1, 3, 4 uploaded as context. Used to ground references to:

- compositionality
- static semantics
- derivation sequences

Lean 4 proofs

- Inspired by *Theorem Proving in Lean 4* (Avigad et al.)
- **All proofs written and verified manually** in VS Code

Next: what retrieval over lecture PDFs looks like in practice.

Querying Lecture Material



Claude generated these lecture summaries by querying Lectures 1, 3, and 4 as project knowledge — each diagram is grounded in the actual PDF content, not general knowledge.

Why Testing Is Not Enough

Dijkstra (1969)

“Testing shows the presence of bugs, not their absence.”

Amazon Rekognition (2018)

28 US Congress members misidentified as criminals after extensive testing — the model was evaluated on the wrong distribution of inputs.

Testing

- Finite set of inputs
- One missed case can fail

Formal proof

- Guaranteed for *all* $x \in \mathbb{R}$
- No exceptions possible

aclu.org/news/privacy-technology/amazons-face-recognition-falsely-matched-28

From Semantics to Lean

Course concept	Lean counterpart
Inference rules (Lec. 3, 4)	Tactics apply rules step-by-step — a derivation tree
Compositionality (Lec. 3)	Proofs compose from sub-proofs, like $\llbracket e_1 + e_2 \rrbracket s$
Static semantics (Lec. 1)	Dependent types enforce constraints at compile time
States: $\text{Var} \rightarrow \mathbb{Z}$ (Lec. 3)	Lean's local context plays the role of the state function

Key idea: course definitions become **checkable proof obligations**. Lec. 3 writes $\mathcal{A}\llbracket a_1 + a_2 \rrbracket s = \mathcal{A}\llbracket a_1 \rrbracket s + \mathcal{A}\llbracket a_2 \rrbracket s$ on paper; Lean enforces it.

Example 1 / 3 ReLU (Rectified Linear Unit) is Non-Negative

relu.lean

```
1 noncomputable def relu (x : R) : R :=  
2   max 0 x  
3  
4 theorem relu_nonneg (x : R) :  
5   relu x >= 0 := by  
6   unfold relu      -- replace name with its body  
7   exact le_max_left 0 x -- close goal: 0 <= max 0 x
```

Why it matters

ReLU is the standard neural network activation function. This proof holds for *all* $x \in \mathbb{R}$ — no test suite can do that.

unfold relu — expands the definition: goal becomes $0 \leq \max 0 x$

exact le_max_left 0 x — applies a Mathlib lemma to close the goal

Course link (Lec. 3): the theorem holds for all $x : \mathbb{R}$ — just as $\mathcal{A}[a]$ is defined for all states s . One proof, infinite coverage.

Example 2 / 3 Vector Dimension Safety

vecadd.lean

```
1  -- n is part of the TYPE, not a runtime variable
2  def vecAdd {n : N} (a b : Fin n -> R)
3    : Fin n -> R :=
4    fun i => a i + b i  -- add element by element
5
6  def v3 : Fin 3 -> R := ![1, 2, 3] -- length 3
7  def v2 : Fin 2 -> R := ![4, 5]    -- length 2
8
9  #check vecAdd v3 v3 -- OK: Fin 3 -> R
10 -- #check vecAdd v3 v2
11 -- ^ TYPE ERROR: Fin 3 /= Fin 2 (compile time)
```

Why it matters

Shape mismatch is the most common runtime crash in deep learning frameworks such as PyTorch and TensorFlow. Lean catches it *before* the program runs — at zero runtime cost.

No theorem to prove — the type system does the work

{n : N} — *n* is implicit: Lean infers the size from the input

Fin n -> R — a vector of exactly *n* reals; length lives in the type

Course link (Lec. 1): static semantics enforces constraints a grammar cannot express. Here the constraint is: both vectors must have the same length *n*.

Violated at compile time, not at runtime.

Example 3 / 3 MSE (Mean Squared Error) Loss is Non-Negative

mse.lean

```
1 -- ys = true labels, ys' = predictions
2 -- goal: sum of squared differences >= 0
3 theorem mse_nonneg {n : N}
4   (ys ys' : Fin n -> R) :
5     0 <= sum i, (ys i - ys' i) ^ 2 := by
6   apply Finset.sum_nonneg
7   -- reduce to: each term >= 0
8   intro i _ -- pick one arbitrary index i
9   exact sq_nonneg _
10  -- any square is >= 0, goal closed
```

apply sum_nonneg — reduces the sum to proving each term ≥ 0

intro i _ — pick an arbitrary index; discard membership proof

exact sq_nonneg — any real square is ≥ 0 ; goal closed

Course link (Lec. 3): proof composes from parts, like $\mathcal{A}[[a_1 + a_2]]s = \mathcal{A}[[a_1]]s + \mathcal{A}[[a_2]]s$. `sq_nonneg` handles one term; `sum_nonneg` lifts it to the whole sum.

Why it matters

MSE is the most common regression loss function. A verified lower bound of 0 eliminates a class of numerical instability bugs that can silently corrupt training.

Try It Yourself

GitHub repository

`github.com/Njoura7/formal-verification-ai`

What is there

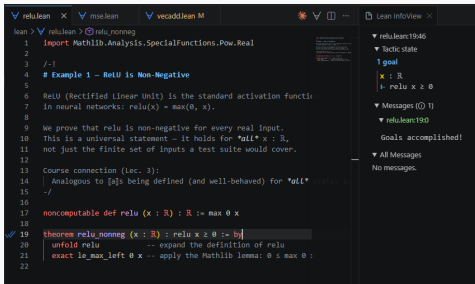
- `lean/relu.lean`
- `lean/vecadd.lean`
- `lean/mse.lean`
- `README.md` with setup steps

To run in VS Code

1. Install `leanprover.lean4` extension
2. `lake exe cache get`
3. Open any `.lean` file
4. See [Goals accomplished!](#)

All three proofs compile against Mathlib — reproducible from a fresh install.

Lean in VS Code



The screenshot shows the Lean IDE in VS Code. The left pane displays a proof file with the following content:

```
lean > rel.lean > rel. nonneg
1  import Mathlib.Analysis.SpecialFunctions.Pow.Real
2
3  /-
4  # Example 1 - ReLU is Non-Negative
5
6  ReLU (Rectified Linear Unit) is the standard activation function
7  in neural networks:  $\text{relu}(x) = \max(0, x)$ .
8
9  We prove that relu is non-negative for every real input.
10 This is a universal statement - it holds for *all*  $x : \mathbb{R}$ ,
11 not just the finite set of inputs a test suite would cover.
12
13 Course connection (lec. 3):
14 Analogous to [a]s being defined (and well-behaved) for *all*  $x : \mathbb{R}$ .
15 -/
16
17 noncomputable def relu (x : ℝ) : ℝ := max 0 x
18
19 theorem relu_nonneg (x : ℝ) : relu x ≥ 0 := by
20   unfold relu -- expand the definition of relu
21   exact le_max_left 0 x -- apply the Mathlib lemma: 0 ≤ max 0 x
22
```

The right pane shows the Lean InfoView for the current goal:

```
▼ rel.lean:19:46
▼ Tactic state
1 goal
x : ℝ
⊢ relu x ≥ 0
▼ Messages (0 / 1)
▼ rel.lean:19:0
Goals accomplished!
▼ All Messages
No messages.
```

What you see

Left — proof file with tactics

Right — Lean Infoview:

- Tactic state shows the open goal
- $\text{relu } x \geq 0$ waiting to be closed
- **Goals accomplished!** once all tactics apply

When AI Starts Proving Things

LeanDojo

NeurIPS 2023

Treats Lean's proof state as an environment. A model learns to select the next tactic — a derivation sequence, step by step.

AlphaProof

DeepMind, 2024

Formalised IMO problems in Lean and trained on proof search. Solved **4 of 6** problems at silver medal level.

DeepSeek-Prover-V2

2025

Splits goals into sub-goals before proving — the same compositional breakdown as denotational semantics.

*These systems do not reason in prose — they work inside a formal language where every step is checkable. **The semantics is the guarantee.***

Conclusion

- Testing is **incomplete** — a proof is not
- Lean tactics *are* derivation sequences from the course
- ReLU, MSE, and shape safety are **formally provable** today
- AI systems are beginning to find those proofs themselves

Takeaway

AI systems can move from *working* to **provably correct**.

References i



Yang, K. et al. *LeanDojo: Theorem Proving with Retrieval-Augmented Language Models*. NeurIPS 2023. [arXiv:2306.15626](https://arxiv.org/abs/2306.15626)



DeepMind. *AI achieves silver-medal standard solving IMO problems*. Blog post, July 2024. deepmind.google



Ying, H. et al. *DeepSeek-Prover-V2*. [arXiv:2504.21801](https://arxiv.org/abs/2504.21801), 2025.



ACLU. *Amazon's Face Recognition Falsely Matched 28 Members of Congress*. 2018. aclu.org



Horpácsi, D. *Formal Semantics — Lecture Notes*, Lec. 1, 3, 4. ELTE, Spring 2026.



Avigad, J. et al. *Theorem Proving in Lean 4*. leanprover.github.io

Questions?